

A LJUNGGREN–NAGELL REDUCTION

ABSTRACT. We prove that the Diophantine equation

$$y(x^3 - z^2) = x$$

has no solution in positive integers. More precisely, we prove that the only solution with $x, y \geq 1$ and $z \geq 0$ is $(x, y, z) = (1, 1, 0)$. The elementary part of the argument reduces the problem to the classical Ljunggren–Nagell input that the equation $U^2 + 1 = A^4 B^3$ has no positive solution. All reductions after that input are given in full detail.

1. THE EXTERNAL INPUT

We use the following standard Ljunggren–Nagell theorem. It is the only external ingredient in the proof.

Theorem 1.1 (Ljunggren–Nagell input). *The equation*

$$U^2 + 1 = A^4 B^3$$

has no solution in integers U, A, B with $U \geq 1$ and $A, B \geq 1$. Equivalently, the only solution with $U \geq 0$ and $A, B \geq 1$ is

$$(U, A, B) = (0, 1, 1).$$

Remark 1.2. The rest of the manuscript is self-contained once Theorem 1.1 is accepted. Appendix A records the Gaussian-integer form of Theorem 1.1, because it explains exactly why this is the natural classical input for the present equation.

2. ELEMENTARY REDUCTIONS

We shall repeatedly use the following elementary fact.

Lemma 2.1 (Coprime factors of a square). *Let $r, s \in \mathbb{Z}_{>0}$ satisfy $\gcd(r, s) = 1$. If rs is a square, then both r and s are squares.*

Proof. Write prime factorizations

$$r = \prod_p p^{\alpha_p}, \quad s = \prod_p p^{\beta_p},$$

where all but finitely many exponents are zero. The condition $\gcd(r, s) = 1$ says that for each prime p , at most one of α_p, β_p is nonzero. Since rs is a square, each $\alpha_p + \beta_p$ is even. Therefore the nonzero exponent among α_p, β_p , if there is one, is even. Thus every prime exponent in r and in s is even, so both r and s are squares. $\square$

Lemma 2.2 (The quotient forced by the equation). *Let $x, y, z \in \mathbb{Z}$ satisfy*

$$x \geq 1, \quad y \geq 1, \quad z \geq 0, \quad y(x^3 - z^2) = x.$$

Then $y \mid x$. If $d = x/y$, then

$$(2.1) \quad z^2 = d(d^2 y^3 - 1).$$

Moreover, if $z > 0$, then $d^2 y^3 - 1 > 0$.

Proof. The defining equation gives

$$x^3 - z^2 = \frac{x}{y}.$$

The left-hand side is an integer. Hence x/y is an integer, so $y \mid x$. Put

$$d = \frac{x}{y} \in \mathbb{Z}_{>0}.$$

Then $x = dy$, and substitution gives

$$y((dy)^3 - z^2) = dy.$$

Dividing by $y > 0$ yields

$$d^3y^3 - z^2 = d,$$

so

$$z^2 = d^3y^3 - d = d(d^2y^3 - 1),$$

which is (2.1). If $z > 0$, then $z^2 > 0$, and since $d > 0$, the factor $d^2y^3 - 1$ must also be positive. $\square$

Lemma 2.3 (The square quotient reduction). *Let $x, y, z \in \mathbb{Z}$ satisfy*

$$x \geq 1, \quad y \geq 1, \quad z \geq 0, \quad y(x^3 - z^2) = x.$$

Then either $(x, y, z) = (1, 1, 0)$, or there are integers $a, u \geq 1$ such that

$$(2.2) \quad x = a^2y, \quad z = au, \quad u^2 + 1 = a^4y^3.$$

Proof. By Lemma 2.2, $d = x/y$ is a positive integer and

$$z^2 = d(d^2y^3 - 1).$$

First suppose

$$d^2y^3 - 1 = 0.$$

Then $d^2y^3 = 1$. Since $d, y \in \mathbb{Z}_{>0}$, this forces $d = 1$ and $y = 1$. Thus $x = dy = 1$, and the displayed formula for z^2 gives $z = 0$. Hence $(x, y, z) = (1, 1, 0)$.

It remains to consider the case

$$d^2y^3 - 1 > 0.$$

The two positive factors d and $d^2y^3 - 1$ are coprime, because any common divisor divides both d and $d^2y^3 - 1$, hence divides

$$d^2y^3 - (d^2y^3 - 1) = 1.$$

Their product is the square z^2 . Lemma 2.1 therefore implies that both factors are squares. Hence there are $a, u \in \mathbb{Z}_{>0}$ such that

$$d = a^2, \quad d^2y^3 - 1 = u^2.$$

The first equality gives

$$x = dy = a^2y.$$

The second gives

$$u^2 + 1 = d^2y^3 = a^4y^3,$$

which is (2.2). Finally,

$$z^2 = d(d^2y^3 - 1) = a^2u^2,$$

and since $z, a, u \geq 0$, with $a, u > 0$, we get $z = au$. $\square$

3. THE SOLUTION OF THE ORIGINAL EQUATION

Theorem 3.1. *The only integer solution of*

$$y(x^3 - z^2) = x$$

with $x, y \geq 1$ and $z \geq 0$ is

$$(x, y, z) = (1, 1, 0).$$

Proof. Let x, y, z be a solution with $x, y \geq 1$ and $z \geq 0$. By Lemma 2.3, either $(x, y, z) = (1, 1, 0)$, or there are integers $a, u \geq 1$ such that

$$u^2 + 1 = a^4y^3.$$

The latter is impossible by Theorem 1.1, applied with

$$U = u, \quad A = a, \quad B = y.$$

Therefore only the first alternative can occur, and the only solution is $(1, 1, 0)$. $\square$

Corollary 3.2. *The equation*

$$y(x^3 - z^2) = x$$

has no solution in positive integers x, y, z .

Proof. A positive-integer solution would in particular be a solution with $x, y \geq 1$ and $z \geq 0$. Theorem 3.1 says that the only such solution is $(1, 1, 0)$, but this has $z = 0$, not $z > 0$. Hence no positive-integer solution exists. $\square$

Remark 3.3. The theorem above proves nonexistence in the positive range and identifies the single boundary solution with $z = 0$. It does not claim a classification when x, y, z are allowed to be arbitrary signed integers.

APPENDIX A. GAUSSIAN-INTEGER FORM OF THE QUOTED INPUT

This appendix is included to make the role of the classical input transparent. It is not a replacement for Theorem 1.1; rather, it shows why the input is exactly the Ljunggren–Nagell obstruction naturally produced by the present Diophantine equation.

Recall that $\mathbb{Z}[i]$ is a unique factorization domain with units $\{\pm 1, \pm i\}$ and norm

$$N(r + si) = r^2 + s^2.$$

Proposition A.1. *Let $U, A, B \in \mathbb{Z}$ satisfy $U \geq 1$ and $A, B \geq 1$. If*

$$(A.1) \quad U^2 + 1 = A^4 B^3,$$

then U is even, A and B are odd, and there exist Gaussian integers $\alpha, \beta \in \mathbb{Z}[i]$ and a unit $\varepsilon \in \{\pm 1, \pm i\}$ such that

$$(A.2) \quad U + i = \varepsilon \alpha^4 \beta^3, \quad N(\alpha) = A, \quad N(\beta) = B.$$

Consequently Theorem 1.1 is equivalent to the assertion that no nonunit Gaussian data in (A.2) can have imaginary part 1.

Proof. First consider parity. If U is odd, then

$$U^2 + 1 \equiv 2 \pmod{4}.$$

The right-hand side $A^4 B^3$ is either odd, if A and B are both odd, or is divisible by 8 if B is even, or by 16 if A is even. It is never congruent to 2 (mod 4). Thus U is even. Then $U^2 + 1$ is odd, so $A^4 B^3$ is odd, and therefore A and B are odd.

In $\mathbb{Z}[i]$ we have

$$(A.3) \quad (U + i)(U - i) = U^2 + 1 = A^4 B^3.$$

We claim that $U + i$ and $U - i$ are coprime in $\mathbb{Z}[i]$. Indeed, any common divisor divides their difference $2i$. Hence the norm of any common divisor is a divisor of $N(2i) = 4$. On the other hand, the norm of any divisor of $U + i$ divides $N(U + i) = U^2 + 1$, which is odd. Therefore the norm of a common divisor is 1, so the common divisor is a unit.

Since $\mathbb{Z}[i]$ is a unique factorization domain and the two factors in (A.3) are coprime, every Gaussian prime occurring in $U + i$ occurs there with exactly the exponent with which its rational norm contributes to $A^4 B^3$. Thus the prime factorization of $U + i$ can be grouped into a fourth-power part, a third-power part, and a unit. Concretely, there are $\alpha, \beta \in \mathbb{Z}[i]$ and a unit ε such that

$$U + i = \varepsilon \alpha^4 \beta^3.$$

Taking norms gives

$$U^2 + 1 = N(\alpha)^4 N(\beta)^3.$$

The grouping may be chosen so that

$$N(\alpha) = A, \quad N(\beta) = B,$$

because the rational prime exponents belonging to A are placed in α , while those belonging to B are placed in β .

Finally, $U + i$ has imaginary part 1, so (A.2) says that a Gaussian product of a fourth power and a third power has imaginary part 1. Conversely, any identity of the form (A.2) with imaginary part 1 gives, after taking norms, an equation of the form (A.1). This proves the claimed equivalence. $\square$